Lecture 12

States that stay and states that leave: decay

Objectives. *For a potential, a state that moves like a free packet at late times leaves every bounded region. By a cited theorem so does every state orthogonal to the bound states. We prove that the survival probability of a state of finite energy spread starts as* $1-(\Delta H)^2t^2/\hbar^2$. *In the Friedrichs model we compute its resolvent exactly, and a cited theorem gives its decay rate.*

- Distinguish survival from remaining in a bounded region.
- Derive the short-time survival law for a unit vector in $D(H)$.
- State the analytic and small-coupling assumptions of the golden rule.

Lecture 12, first part

What decay means for a state

12.1 The point part and the continuous part

Hydrogen atoms in the excited $2p$ level decay by emitting light, with a lifetime of about 1.6 ns (the physics box of § 12.5). In this lecture we ask what decay means for a state and its evolution, in models simpler than an atom coupled to light. We first test whether the state is still the prepared one.

The return to a prepared ray is tested by the projection $|\psi\rangle\langle\psi|$, with ψ a unit vector. After a yes the state lies in this ray. Other nonorthogonal states can also give yes, so the reading does not identify the pre-reading preparation. By Postulate 5.6, on the evolved state $U_t\psi$ it answers yes with the probability $\|\langle\psi, U_t\psi\rangle\,\psi\|^2$. For a unit vector ψ we call

$$\langle\psi, U_t\psi\rangle \quad \text{and} \quad \mathcal{P}(t) = |\langle\psi, U_t\psi\rangle|^2$$

the *survival amplitude* and the *survival probability*. An eigenvector only changes its phase, and so survives with probability 1.

Definition 12.1 (Point part and continuous part)**.** Let H be self-adjoint. The *point part* $\mathcal{H}_{\mathrm{pp}}$ of H is the closure of the set of finite sums of eigenvectors of H. The *continuous part* $\mathcal{H}_{\mathrm{c}}$ is the set of vectors orthogonal to every vector of $\mathcal{H}_{\mathrm{pp}}$.

By the projection step of the proof of Lemma 6.5, every vector is $\psi = \psi_{\mathrm{pp}} + \psi_{\mathrm{c}}$, with ψ_{pp} in the point part and ψ_{c} in the continuous part. Eigenvectors for different eigenvalues are orthogonal, since H is symmetric. Choosing an orthonormal basis of each eigenspace gives an orthonormal family (φ_k) with $H\varphi_k = E_k\varphi_k$, and every vector of $\mathcal{H}_{\mathrm{pp}}$ is a convergent sum $\sum_k c_k\varphi_k$ with $c_k = \langle\varphi_k, \psi\rangle$. For the box and the oscillator, $\mathcal{H}_{\mathrm{pp}}$ is the whole space. For

the free particle, which has no eigenvalue (Theorem 4.6), it is $\{0\}$. For a square well it is spanned by the finitely many bound states of Theorem 10.8.

Proposition 12.2 (The point part). *Let (U_t) be a strongly continuous unitary group with the Hamiltonian H.*

1. *If $H\varphi = E\varphi$, then $U_t\varphi = e^{-\mathrm{i}Et/\hbar}\varphi$ for every t. Each U_t maps $\mathcal{H}_{\mathrm{pp}}$ onto $\mathcal{H}_{\mathrm{pp}}$ and $\mathcal{H}_{\mathrm{c}}$ onto $\mathcal{H}_{\mathrm{c}}$.*
2. *Let $\psi = \sum_k c_k\varphi_k$ be a unit vector of $\mathcal{H}_{\mathrm{pp}}$, and for each eigenvalue E let w_E be the sum of the $|c_k|^2$ with $E_k = E$. Then*

$$\langle\psi, U_t\psi\rangle = \sum_E w_E\, e^{-\mathrm{i}Et/\hbar} \quad \text{and} \quad \frac{1}{T}\int_0^T \mathcal{P}(t)\,\mathrm{d}t \longrightarrow \sum_E w_E^2 > 0 \quad \text{as } T \to \infty.$$

In particular, $\langle\psi, U_t\psi\rangle$ does not tend to 0.

Proof.

1. Since $\varphi \in D(H)$, Proposition 7.3 gives $U_t\varphi$ the derivative $-(\mathrm{i}/\hbar)U_tH\varphi = -(\mathrm{i}E/\hbar)U_t\varphi$. So $e^{\mathrm{i}Et/\hbar}U_t\varphi$ has the derivative 0. Its inner product with every vector is then constant, so it is constant, equal to φ. Hence U_t maps finite sums of eigenvectors to such sums, and, being bounded, it maps $\mathcal{H}_{\mathrm{pp}}$ into $\mathcal{H}_{\mathrm{pp}}$. So does $U_{-t} = U_t^\dagger$. If $\chi \in \mathcal{H}_{\mathrm{c}}$ and $\eta \in \mathcal{H}_{\mathrm{pp}}$, then $\langle\eta, U_t\chi\rangle = \langle U_{-t}\eta, \chi\rangle = 0$, so U_t maps $\mathcal{H}_{\mathrm{c}}$ into $\mathcal{H}_{\mathrm{c}}$. Since $U_tU_{-t} = \mathbb{1}$, both maps are onto.
2. Since U_t is bounded, the first claim gives $U_t\psi = \sum_k c_k e^{-\mathrm{i}E_kt/\hbar}\varphi_k$, and the inner product with ψ is $\sum_k |c_k|^2\, e^{-\mathrm{i}E_kt/\hbar}$. Grouping equal E_k gives the first formula, and $\mathcal{P}(t) = \sum_{E,E'} w_E w_{E'} e^{-\mathrm{i}(E-E')t/\hbar}$, a double series that converges uniformly in t, since $\sum_E w_E = 1$. Its mean over $[0,T]$ is the series of the means. The mean of $e^{-\mathrm{i}(E-E')t/\hbar}$ is 1 if $E = E'$, and otherwise has modulus at most $2\hbar/(|E-E'|\,T)$. Given $\epsilon > 0$, choose finitely many levels whose weights sum to more than $1-\epsilon$. The terms with a level outside them contribute at most 2ϵ, and the finitely many others have a mean that tends to $\sum_E w_E^2$ up to 2ϵ. Some w_E is positive, so the limit is positive. If $\langle\psi, U_t\psi\rangle$ tended to 0, so would the mean of $\mathcal{P}$. ∎

A unit vector of the point part has a survival probability whose long-time mean is $\sum_E w_E^2$. For an eigenvector this is 1. The continuous part holds what the eigenvectors miss, and § 12.2 shows that for a potential its states leave. As Lecture 6 announced, the two parts sort states by their distributions μ_ψ of energy readings (5.1). By the spectral theorem, $E_H(\{E\})$ is the projection onto the eigenspace of E (Teschl, p. 112, cited), so ψ lies in $\mathcal{H}_{\mathrm{pp}}$ exactly when μ_ψ is a sum of point weights, as for an eigenbasis in Proposition 5.3, and in $\mathcal{H}_{\mathrm{c}}$ exactly when μ_ψ gives no single energy a positive weight. A distribution without point weights need not have a density. By the spectral theorem, $\mathcal{H}_{\mathrm{c}}$ is the orthogonal sum of an *absolutely continuous* part, whose states have distributions with a density, and a *singular continuous* part (Teschl, Lemma 3.19, cited). For a potential the singular continuous part is $\{0\}$ (Teschl, Thm 12.11, cited).

Self-check 12.1

1. For the box, let $\psi = (u_1 + u_2)/\sqrt{2}$. Find $\mathcal{P}(t)$ and its mean over long times. At which times is $\mathcal{P}(t) = 1$?

12.2 States that leave

A free state has no point part. Two questions follow: does it become orthogonal to its start, and does it leave every bounded region? The first is answered by the Riemann–Lebesgue lemma.

Lemma 12.3 (Riemann–Lebesgue). *For every $\psi \in L^2(\mathbb{R})$,*

$$\int_{-\infty}^{\infty} e^{-\mathrm{i}tx/\hbar}\,|\psi(x)|^2\,\mathrm{d}x \longrightarrow 0 \quad \text{as } t \to \pm\infty.$$

Proof. Let $\epsilon > 0$. By S4 there is a smooth χ that vanishes outside a bounded interval, with $\|\psi - \chi\| < \epsilon$. Since $\left||\psi|^2 - |\chi|^2\right| \le |\psi - \chi|\,(|\psi| + |\chi|)$, the Cauchy–Schwarz inequality gives $\int \left||\psi|^2 - |\chi|^2\right| \mathrm{d}x \le \epsilon(2\,\|\psi\| + \epsilon)$. For χ, an integration by parts with no boundary terms gives

$$\int e^{-\mathrm{i}tx/\hbar}\,|\chi|^2\,\mathrm{d}x = \frac{\hbar}{\mathrm{i}t}\int e^{-\mathrm{i}tx/\hbar}\,(|\chi|^2)'\,\mathrm{d}x,$$

which tends to 0, since $(|\chi|^2)'$ is continuous and vanishes outside a bounded interval, so the last integral is bounded. So the limit superior of the modulus of the integral is at most $\epsilon(2\,\|\psi\| + \epsilon)$, for every $\epsilon > 0$. ∎

Every probability density w on the line is $|\psi|^2$ for $\psi = \sqrt{w}$, so the lemma says that $\int e^{-\mathrm{i}tE/\hbar} w(E)\,\mathrm{d}E \to 0$. In general, for a state whose distribution μ_ψ has the density w, the survival amplitude is $\int e^{-\mathrm{i}tE/\hbar} w(E)\,\mathrm{d}E$ (Hall's functional calculus after Theorem 7.5, cited), and it tends to 0.

The second needs the decay step of Theorem 11.7.

Proposition 12.4 (Free states leave). *Let $\chi \in L^2(\mathbb{R})$ and $L > 0$. As $t \to \pm\infty$, $\left\|1_{[-L,L]} U_t^0 \chi\right\| \to 0$, and $\langle \varphi, U_t^0 \chi\rangle \to 0$ for every φ.*

Proof. The operators $1_{[-L,L]}U_t^0$ have norm at most 1, so it suffices to prove the first claim for χ in a dense set. Take the sums of the vectors χ whose transform is smooth and vanishes outside $[a,b]$ or outside $[-b,-a]$, with $0 < a < b$, as in the proof of Theorem 11.7. Its decay step, on $[-L,L]$, gives $\left|(U_t^0\chi)(x)\right| \le C/t^2$ there for large $|t|$, so $\left\|1_{[-L,L]}U_t^0\chi\right\| \le (2L)^{1/2}C/t^2$. For the second claim, split $\varphi = 1_{[-L,L]}\varphi + 1_{|x|>L}\varphi$. Since $1_{[-L,L]}$ is self-adjoint,

$$\left|\langle \varphi, U_t^0\chi\rangle\right| \le \|\varphi\|\,\left\|1_{[-L,L]}U_t^0\chi\right\| + \left\|1_{|x|>L}\varphi\right\|\,\|\chi\|\,.$$

The first term tends to 0, and by the continuity clause of Definition 5.1 for the projections 1_I the second is small for large L. ∎

So a state that leaves every bounded region becomes orthogonal to every fixed vector, in particular to its start. How a free state leaves is a cited theorem.

Theorem 12.5 (Late times · *cited*). *Let $\psi \in L^2(\mathbb{R})$ be a unit vector, and let I be an interval. As $t \to \infty$,*

$$\mathrm{Prob}_{U_t^0\psi}(Q \in tI) - \mathrm{Prob}_{\psi}(P \in mI) \longrightarrow 0,$$

where tI and mI are the intervals of the numbers tv and mv, $v \in I$.

This follows from Amrein, Prop. 5.18, which writes $(U_t^0)^\dagger E_Q(tI) U_t^0$ as the projection $E_P(mI)$ conjugated by a unitary Z_t that tends to $\mathbb{1}$ (his $\hbar = 1$ and $m = \frac{1}{2}$), and from Teta, Prop. 6.2.

Thus probabilities of the scaled position Q/t on intervals tend to those of the velocity P/m. This is a limit of distributions, not an equality of densities at a finite time. For the Gaussian packet (4.3), the mean of Q/t in $U_t^0\psi$ is $p_0/m + q/t$, and by (7.6) its spread $\Delta Q_t/t$ tends to $\hbar/2ms = \Delta P/m$.

This law gives a way to read momentum, called *time of flight*, as in Example 7.13. Let a particle in the box have the state ψ, and let φ be ψ extended by zero to the line. Suddenly remove the walls at $t = 0$, and assume that the particle then evolves freely. By Theorem 12.5, for every interval I,

$$\mathrm{Prob}_{U_t^0\varphi}(mQ/t \in I) \longrightarrow \int_I |\tilde{\varphi}(p)|^2 \, \mathrm{d}p \quad \text{as } t \to \infty.$$

So the position read at a late time, multiplied by m/t, is distributed nearly as the momentum of φ. This is a limit of distributions. It is not a formula for a detector at a finite time, and it is not a reading of any P_θ of the intact box. For $\psi = u_1$, the limiting density is $(L/2\pi\hbar)\, y(pL/\hbar)$, with the function y of Figure 3.2.

With a potential, the states split. Let V be a potential (Definition 10.1), $H = H_0 + V$, and (U_t) its group (Proposition 11.6).

Theorem 12.6 (States that stay and states that leave). *Let V be a potential.*

1. *If $\psi \in \mathcal{H}_{\mathrm{pp}}$, then for every $\epsilon > 0$ there is L with $\left\|1_{|x|>L} U_t \psi\right\| \le \epsilon$ for every t.*
2. *If $\psi = \Omega_{\mathrm{out}}\chi$, then for every $L > 0$, $\left\|1_{[-L,L]} U_t \psi\right\| \to 0$ and $\langle \varphi, U_t\psi\rangle \to 0$ for every φ, as $t \to \infty$. The same holds for $\psi = \Omega_{\mathrm{in}}\chi$ as $t \to -\infty$.*

Proof.

1. Write $\psi = \sum_k c_k \varphi_k$ as in Proposition 12.2, and choose a finite set K of indices such that the sum ψ_K of the terms with $k \in K$ has $\|\psi - \psi_K\| \le \epsilon/2$. By Proposition 12.2, $U_t\psi_K = \sum_{k\in K} c_k e^{-\mathrm{i}E_k t/\hbar}\varphi_k$, and U_t keeps norms, so

$$\left\|1_{|x|>L} U_t\psi\right\| \le \frac{\epsilon}{2} + \sum_{k\in K} |c_k| \, \left\|1_{|x|>L}\varphi_k\right\|$$

   for every t. By the continuity clause of Definition 5.1, the finite sum is at most $\epsilon/2$ for large L.

2. Since U_t is unitary, $\left\|U_t\psi - U_t^0\chi\right\| = \left\|\Omega_{\mathrm{out}}\chi - U_t^\dagger U_t^0\chi\right\| \to 0$ as $t \to \infty$ (Theorem 11.7). So

$$\left\|1_{[-L,L]} U_t\psi\right\| \le \left\|1_{[-L,L]} U_t^0\chi\right\| + \left\|U_t\psi - U_t^0\chi\right\|,$$

   which tends to 0 by Proposition 12.4. The second claim follows from the first as in the proof of Proposition 12.4. ∎

By Theorem 11.7, the ranges of the wave operators lie in $\mathcal{H}_{\mathrm{c}}$. Enss's theorem, cited in § 11.5, says that they fill it.

Corollary 12.7 (The continuous part leaves · *sketched*). *Let V be a potential and $\psi \in \mathcal{H}_{\mathrm{c}}$. For every $L > 0$, $\left\|1_{[-L,L]} U_t\psi\right\| \to 0$ as $t \to \infty$ and as $t \to -\infty$, and $\langle \varphi, U_t\psi\rangle \to 0$ for every φ.*

Proof. By Enss's theorem (Teschl, Thm 12.12, cited), ψ is some $\Omega_{\mathrm{out}}\chi$ and some $\Omega_{\mathrm{in}}\chi'$. Apply Theorem 12.6. ∎

So the two parts of a state behave in two ways. With $\psi = \psi_{\mathrm{pp}} + \psi_{\mathrm{c}}$, at late times a reading of the position in $[-L, L]$ sees only the point part. Its probability $\left\| 1_{[-L,L]} U_t \psi \right\|^2$ differs from $\left\| 1_{[-L,L]} U_t \psi_{\mathrm{pp}} \right\|^2$ by an amount that tends to 0. For a square well with a single level and ground state φ, the probability of a reading in $[-L, L]$ tends to $|\langle \varphi, \psi \rangle|^2 \left\| 1_{[-L,L]} \varphi \right\|^2$, close to $|\langle \varphi, \psi \rangle|^2$ for large L. So the particle stays near the well with this probability and leaves with the rest.

Figure 12.1 shows this for one preparation, in units with $m = \hbar = 1$. The well is $V = -0.72$ on $|x| < 1$. In the notation of Theorem 10.8, $\gamma = 1.2$, so it has one level. Its bound state is $\varphi(x) = A \cos(kx)$ for $|x| \le 1$ and $\varphi(x) = A \cos(k) e^{-\eta(|x|-1)}$ for $|x| \ge 1$, with $k \approx 0.8193$, $\eta \approx 0.8768$ and $A \approx 0.6835$. The state is $\psi(x) = (2\pi)^{-1/4} e^{-x^2/4 + ix}$, with $|\langle \varphi, \psi \rangle|^2 \approx 0.4034$. For $L = 4$ the limit is about 0.4029, not 0.4034, since φ has tails outside $[-4, 4]$. The figure is a numerical evolution of this one state: it illustrates the limit and does not prove it.

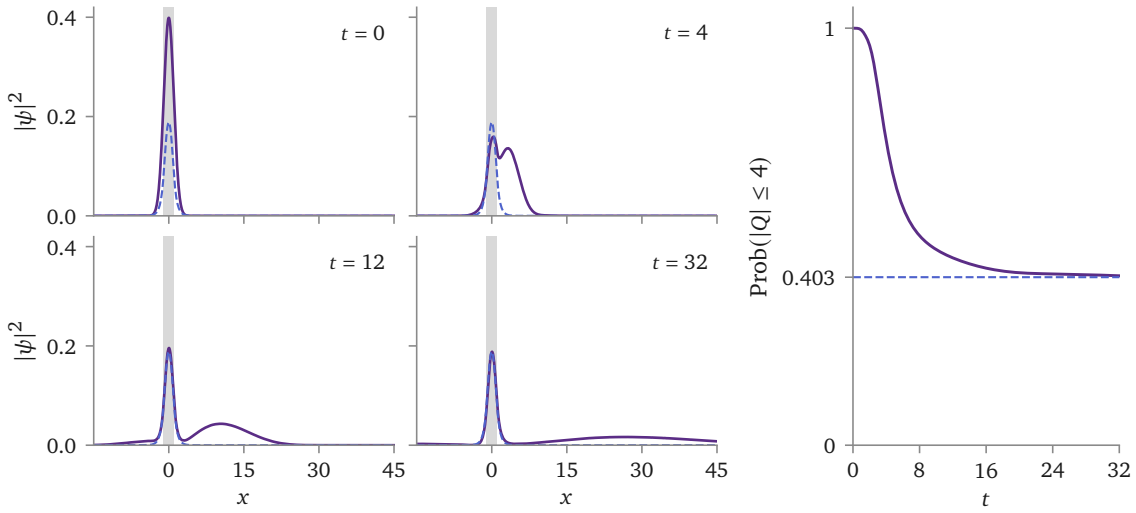


Figure 12.1: The state ψ in the square well of the text, evolved by a unitary numerical method, the split-step Fourier method on $[-256, 256)$, with grid spacing $1/32$ and time step 0.005; the norm stays 1 to within 10^{-12}. Left: the density $|U_t \psi|^2$ at $t = 0, 4, 12, 32$, with the well grey and $|\langle \varphi, \psi \rangle|^2 |\varphi|^2$ dashed. Right: the probability of a reading in $[-4, 4]$ against t, with the limit 0.4029 dashed.

The same state has a distribution of energies with both parts, as Example 5.2 announced. The well has one level and no eigenvalue ≥ 0 (Theorem 10.7), and for a potential the singular continuous part is $\{0\}$ (§ 12.1). So μ_ψ is the point weight $|\langle \varphi, \psi \rangle|^2 \approx 0.4034$ at the level $E_1 = -\hbar^2 \eta^2 / 2m \approx -0.3844$, plus a density w on $(0, \infty)$ that carries the rest, about 0.5966. The point weight is $\left\| \psi_{\mathrm{pp}} \right\|^2$, the probability that stays near the well, and the continuous part carries $\|\psi_{\mathrm{c}}\|^2$, the probability that leaves. Figure 12.2 draws both.

The RAGE theorem (Teschl, Thms 5.6–5.7, cited) needs no asymptotic completeness and holds in three dimensions too. For every Hamiltonian with the local compactness of Lemma 10.9, a state of $\mathcal{H}_{\mathrm{c}}$ leaves every bounded region on time average, in the sense that

$$\frac{1}{T} \int_0^T \left\| 1_{[-L,L]} U_t \psi \right\|^2 \, \mathrm{d}t \longrightarrow 0 \quad \text{as } T \to \infty.$$

For sequences with bounded norms, the lemma, stated for H_0, carries over to $H = H_0 + V$, since $\|(H_0 - z)\psi\| \le \|(H - z)\psi\| + \sup |V| \, \|\psi\|$.

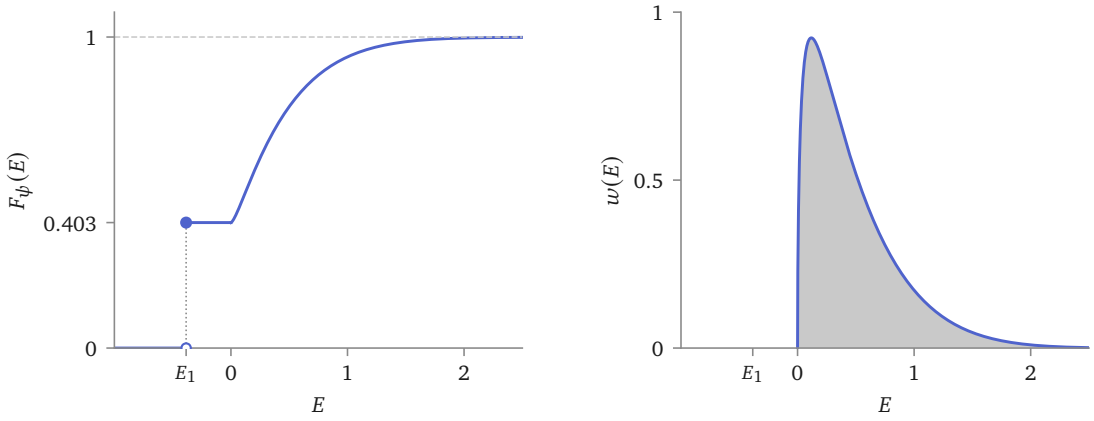


Figure 12.2: The distribution of energies of the state ψ of Figure 12.1, in units with $m = \hbar = 1$. Left: the distribution function F_ψ, which jumps by 0.4034 at the level E_1, is constant on $[E_1, 0]$, and then rises continuously towards 1. Right: the density w of the continuous part, the slope of F_ψ on $(0, \infty)$, of area 0.5966. It is computed as the limit, as $\epsilon \to 0$, of $(1/\pi)\,\mathrm{Im}\,\langle \psi, (H - E - \mathrm{i}\epsilon)^{-1}\psi\rangle$ (Teschl, Thm 3.23, cited in § 12.5), not at a fixed $\epsilon > 0$: there this function also contains the point weight, spread into a Lorentzian about E_1.

Local compactness is needed. Let $H = Q$, so that $U_t\psi = e^{-\mathrm{i}tx/\hbar}\psi$ (Theorem 7.4). It has no eigenvalue, and by Lemma 12.3 the survival amplitude $\int e^{-\mathrm{i}tx/\hbar}\,|\psi|^2\,\mathrm{d}x$ tends to 0. Yet $|U_t\psi| = |\psi|$. So the position density never changes, and nothing leaves. The orthonormal vectors $\psi_n(x) = e^{2\pi\mathrm{i}nx}1_{[0,1]}(x)$ lie in $[-1, 1]$ and have $\|Q\psi_n\| \le 1$, yet no subsequence of them converges, since $\|\psi_n - \psi_k\|^2 = 2$ for $n \neq k$. So Q is not locally compact. Becoming orthogonal to its start is not the same as leaving.

Self-check 12.2

1. Let $H = Q$ and $\psi = 1_{[0,1]}$. Compute $\langle \psi, U_t\psi\rangle$, and show that its modulus is at most $2\hbar/|t|$.

12.3 Short times

Decay is often described by $\mathcal{P}(t) = e^{-\Gamma t}$. At short times this law fails for every state of finite energy spread. Recall that $\Delta H = \infty$ for $\psi \notin D(H)$ (Lecture 2).

Proposition 12.8 (Short times). *Let (U_t) be a strongly continuous unitary group with the Hamiltonian H, and let $\psi \in D(H)$ be a unit vector, with $\Delta H = \|H\psi - \langle \psi, H\psi\rangle\,\psi\|$. Then*

$$\mathcal{P}(t) = 1 - \frac{(\Delta H)^2 t^2}{\hbar^2} + o(t^2) \quad \text{as } t \to 0.$$

Proof. Write $a(t) = \langle \psi, U_t\psi\rangle$. Since $U_t\psi$ and ψ are unit vectors and $\langle \psi, U_t\psi\rangle = a(t)$,

$$\|U_t\psi - a(t)\psi\|^2 = 1 - 2\,|a(t)|^2 + |a(t)|^2 = 1 - \mathcal{P}(t).$$

Since $\langle \psi, \psi\rangle = 1$,

$$U_t\psi - a(t)\psi = (U_t\psi - \psi) - \langle \psi, U_t\psi - \psi\rangle\,\psi.$$

By Proposition 7.3, $(U_t\psi - \psi)/t \to -(\mathrm{i}/\hbar)H\psi$. Dividing by t, the right side tends to $-(\mathrm{i}/\hbar)(H\psi - \langle \psi, H\psi\rangle\,\psi)$, whose norm is $\Delta H/\hbar$. So $(1 - \mathcal{P}(t))/t^2 \to (\Delta H)^2/\hbar^2$. ∎

An exponential law $e^{-\Gamma|t|}$ with $\Gamma > 0$ has $(1 - e^{-\Gamma|t|})/t^2 \to \infty$. So no state of finite energy spread decays exactly exponentially, not even on a short interval after $t = 0$. It starts flat. At late times the spectrum decides.

Theorem 12.9 (Khalfin · *cited*)**.** *Let H be self-adjoint and bounded below, (U_t) its group, and ψ a unit vector. There are no numbers C and $\beta > 0$ with $\mathcal{P}(t) \le Ce^{-\beta t}$ for all $t > 0$.*

This is Exner, Thm 1.5.7, for the survival probability of one vector. There a bound $Ce^{-\beta t}$ forces the spectrum to be the whole line. Its proof continues the energy density analytically into a strip of complex numbers (Exner, Lemma 1.5.5), a tool of complex analysis. Khalfin first tied the failure of exponential decay to a spectrum bounded below, in 1957.

The flat start has a consequence for repeated readings. Let $\psi \in D(H)$, and test for the prepared ray at the times $t/n, 2t/n, \dots, t$. Then all n answers are yes with the probability $\mathcal{P}(t/n)^n$, and by Proposition 12.8 it tends to 1 as $n \to \infty$. This limit is the *Zeno effect*. It concerns the system that is tested. After each yes the state is again the ray of ψ, so after the last test, if all answers are yes, a position reading has the distribution of ψ. For the evolution without tests, the limit says nothing: a free state in $D(H_0)$ also starts flat, and it leaves every bounded region (Proposition 12.4).

Self-check 12.3

1. Let $\psi \in D(H)$. Read whether the state is still ψ at the times $t/n, 2t/n, \dots, t$, as at the start of § 12.1. Show that all n answers are yes with the probability $\mathcal{P}(t/n)^n$, and compute the limit of $n \log \mathcal{P}(t/n)$ as $n \to \infty$.

12.4 A state that decays exactly

Both obstacles can be avoided, but only together.

Example 12.10 (The Lorentzian)**.** Let ε_0 be real and $\Gamma > 0$. Let H be multiplication by the variable E on $L^2(\mathbb{R})$, self-adjoint by Lemma 0.7, and let $\psi = \sqrt{\rho}$, where

$$\rho(E) = \frac{\hbar\Gamma/2\pi}{(E - \varepsilon_0)^2 + (\hbar\Gamma/2)^2}$$

is the *Lorentzian* of half-width $\hbar\Gamma/2$ about ε_0, with $\int \rho \, \mathrm{d}E = 1$. By Theorem 7.4, $U_t\psi = e^{-\mathrm{i}tE/\hbar}\psi$, and

$$\langle \psi, U_t\psi \rangle = \int_{-\infty}^{\infty} e^{-\mathrm{i}tE/\hbar}\rho(E)\,\mathrm{d}E = e^{-\mathrm{i}\varepsilon_0 t/\hbar - \Gamma|t|/2}, \qquad \mathcal{P}(t) = e^{-\Gamma|t|}$$

for every t. To see the integral, take its inverse. For every real E,

$$\frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} e^{\mathrm{i}(E-\varepsilon_0)t/\hbar - \Gamma|t|/2}\,\mathrm{d}t = \frac{1}{2\pi\hbar}\left(\frac{1}{\Gamma/2 - \mathrm{i}(E-\varepsilon_0)/\hbar} + \frac{1}{\Gamma/2 + \mathrm{i}(E-\varepsilon_0)/\hbar}\right) = \rho(E),$$

where the half-lines $t > 0$ and $t < 0$ give the two elementary integrals. By the inversion formula of the standing fact S1, the integral of $e^{-\mathrm{i}tE/\hbar}\rho(E)$ is the function of t above.

The state decays exactly exponentially, at the rate Γ, and Proposition 12.8 and Theorem 12.9 do not apply. Since $E^2\rho(E) \to \hbar\Gamma/2\pi$ as $|E| \to \infty$, the integral $\int E^2\rho \, \mathrm{d}E$ diverges, so $\psi \notin D(H)$ and $\Delta H = \infty$. Even $\int |E|\,\rho \, \mathrm{d}E$ diverges, so the state has no mean energy. And H is not bounded below. A Lorentzian distribution of energies gives an exactly exponential decay. By Theorem 12.9, a Hamiltonian bounded below never does, so the decay of a real

system, whose energy is bounded below, can be exponential only approximately. Exner, §3.1, calls replacing a system by this model the *pole-and-real-axis approximation*.

Self-check 12.4

1. Show that $\rho(\varepsilon_0 \pm \hbar\Gamma/2) = \rho(\varepsilon_0)/2$. A level with the lifetime $1/\Gamma = 1\,\text{ns}$ has which half-width, in eV? Use $\hbar = 6.58 \times 10^{-16}\,\text{eV s}$.

So the first part has separated two meanings of decay. For a potential, a state that moves like a free packet at late times leaves every bounded region, and by Enss's cited theorem so does every state of the continuous part. The survival probability of a state can equal $e^{-\Gamma|t|}$, with $\Gamma > 0$, for all t only if the state lies outside $D(H)$ (Proposition 12.8) and H is not bounded below (Theorem 12.9, cited), as for the Lorentzian.

Lecture 12, second part

The decay rate of a level in a continuum

The first part proved that the survival probability of a state in $D(H)$ starts flat. It cited Khalfin's theorem: if H is bounded below, no $Ce^{-\beta t}$ with $\beta > 0$ bounds the survival probability for all $t > 0$. In this part we couple a level to a continuum in the Friedrichs model. We compute its resolvent exactly, quote the theorem that gives its decay rate, the golden rule, and compute the level's distribution of energies and its survival for one example.

12.5 A level in a continuum: the golden rule

An excited atom has a level, but it is coupled to a continuum: the atom in a lower state with a photon of any energy. The simplest model of a level inside a continuum is Friedrichs'. Its space $\mathcal{H} = \mathbb{C} \oplus L^2(0,\infty)$ holds the pairs $\{\alpha, f\}$ of a number and a function of the energy $\lambda > 0$, with $\langle\{\alpha, f\}, \{\beta, h\}\rangle = \overline{\alpha}\beta + \langle f, h\rangle$. Its vector $e = \{1, 0\}$ is the level.

Definition 12.11 (The Friedrichs model). Let $\lambda_0 > 0$, $v \in L^2(0,\infty)$ and g real. On $D(H_0) = \{\{\alpha, f\} : \lambda f \in L^2(0,\infty)\}$ put

$$H_0\{\alpha, f\} = \{\lambda_0\alpha, \lambda f\}, \qquad V\{\alpha, f\} = \{\langle v, f\rangle, \alpha v\}, \qquad H_g = H_0 + gV.$$

Here V is the coupling, not a potential. The operator H_0 multiplies by λ_0 on $\mathbb{C}$ and by λ on $L^2(0,\infty)$, so, by the proofs of Lemma 0.7 and Theorem 7.4, word for word on $\mathbb{C} \oplus L^2(0,\infty)$, it is self-adjoint and generates

$$e^{-\mathrm{i}tH_0/\hbar}\{\alpha, f\} = \{e^{-\mathrm{i}\lambda_0 t/\hbar}\alpha, e^{-\mathrm{i}\lambda t/\hbar} f\}.$$

Its eigenvector e has the eigenvalue λ_0, inside the spectrum $[0,\infty)$ of the continuum. The coupling is bounded, since by the Cauchy–Schwarz inequality

$$\|V\{\alpha, f\}\|^2 = |\langle v, f\rangle|^2 + |\alpha|^2\,\|v\|^2 \le \|v\|^2\,\|\{\alpha, f\}\|^2,$$

and symmetric, since

$$\langle\{\beta, h\}, V\{\alpha, f\}\rangle = \overline{\beta}\,\langle v, f\rangle + \alpha\,\langle h, v\rangle = \langle V\{\beta, h\}, \{\alpha, f\}\rangle.$$

So the proof of Proposition 10.2 shows that H_g is self-adjoint, and the proof of Proposition 11.6, with H_0's group in place of the free one, builds its group (U_t). Since $H_g e = \{\lambda_0, gv\}$,

the level has the mean λ_0 and the spread $\Delta H = |g|\,\|v\|$, and by Proposition 12.8 it starts to decay as $1 - g^2\,\|v\|^2\,t^2/\hbar^2$. Since $\langle\psi, H_0\psi\rangle \geq 0$ and $\|gV\| \leq |g|\,\|v\|$, the means satisfy $\langle\psi, H_g\psi\rangle \geq -|g|\,\|v\|$ for unit $\psi \in D(H_0)$, so H_g is bounded below (Proposition 10.3), and Theorem 12.9 applies.

Proposition 12.12 (The level's resolvent). *For every non-real z,*

$$\langle e, (H_g - z)^{-1} e\rangle = \Big(\lambda_0 - z - g^2 \int_0^\infty \frac{|v(\lambda)|^2}{\lambda - z}\,\mathrm{d}\lambda\Big)^{-1}.$$

Proof. By Theorem 6.3, z lies in the resolvent set. Let $\{\alpha, f\} = (H_g - z)^{-1}e$, so that $\alpha = \langle e, (H_g - z)^{-1}e\rangle$. The two components of $(H_g - z)\{\alpha, f\} = e$ read

$$(\lambda_0 - z)\alpha + g\,\langle v, f\rangle = 1 \quad \text{and} \quad (\lambda - z)f + g\alpha v = 0.$$

The second gives $f = -g\alpha v/(\lambda - z)$, so $\langle v, f\rangle = -g\alpha \int |v|^2/(\lambda - z)\,\mathrm{d}\lambda$, and the first becomes α times the bracket of the claim, equal to 1. ∎

For $g = 0$ the bracket vanishes at λ_0, the eigenvalue of e. A cited theorem says where this zero goes for a small coupling.

A function of a complex variable is *analytic* on an open set if it has a convergent power series about each point of that set. An *analytic continuation* is an analytic function on a larger region that agrees with the original function wherever both are defined. A function is *meromorphic* if it is analytic except at isolated *poles*. Near a pole z_0 it has the form $b(z)/(z - z_0)^j$, where $j \geq 1$ and b is analytic with $b(z_0) \neq 0$. The pole is *simple* when $j = 1$. Continuing from the upper half-plane across a real interval means extending the same analytic function through that interval to points below it. No proof using complex analysis is assumed here.

Theorem 12.13 (The golden rule · *cited*). *In the Friedrichs model, suppose that $|v|^2$ extends from $(0, \infty)$ to a meromorphic function on an open set of complex numbers that contains $(0, \infty)$, and that $v(\lambda_0) \neq 0$. Then for every nonzero real g small enough, the function $z \mapsto \langle e, (H_g - z)^{-1}e\rangle$, continued from the upper half-plane across $(0, \infty)$, has a simple pole at a point $E_{\mathrm{res}} - \mathrm{i}\hbar\Gamma/2$ with*

$$\Gamma = \frac{2\pi g^2\,|v(\lambda_0)|^2}{\hbar} + O(g^4).$$

This is Exner, Thm 3.2.3 with (2.8b) and (2.9a), in units with $\hbar = 1$ and without his diagonal term. The pole is a *resonance*. It belongs to the continued scalar function. It is not a nonreal eigenvalue of H_g. The actual resolvent of the self-adjoint H_g exists at nonreal z, and its values below the real axis need not equal this continuation. The next example shows a continued function with a pole beside an actual resolvent without one, in a model where every formula is explicit.

Example 12.14 (Two branches of one resolvent). Take the Lorentzian state ψ of Example 12.10, put $a = \hbar\Gamma/2$, and let

$$R(z) = \langle\psi, (H - z)^{-1}\psi\rangle = \int_{-\infty}^{\infty} \frac{\rho(E)}{E - z}\,\mathrm{d}E$$

for nonreal z. Let $\operatorname{Im} z > 0$. Then $(E - z)^{-1} = (\mathrm{i}/\hbar)\int_0^\infty e^{-\mathrm{i}t(E-z)/\hbar}\,\mathrm{d}t$, and the double integral of $\rho(E)e^{-t\operatorname{Im} z/\hbar}$ is finite, so the two integrations can be exchanged. The survival amplitude

of Example 12.10 then gives

$$R(z) = \frac{\mathrm{i}}{\hbar}\int_0^\infty e^{\mathrm{i}tz/\hbar}\,\langle\psi, U_t\psi\rangle\,\mathrm{d}t = \frac{\mathrm{i}}{\hbar}\int_0^\infty e^{\mathrm{i}t(z-\varepsilon_0)/\hbar-\Gamma t/2}\,\mathrm{d}t = \frac{1}{\varepsilon_0 - z - \mathrm{i}a}.$$

Since ρ is real, $R(\bar{z}) = \overline{R(z)}$, so $R(z) = (\varepsilon_0 - z + \mathrm{i}a)^{-1}$ for $\operatorname{Im} z < 0$. The upper formula is analytic at every $z \neq \varepsilon_0 - \mathrm{i}a$. So it continues R from the upper half-plane across the real axis, and the continuation has a simple pole at $\varepsilon_0 - \mathrm{i}a$, at the distance $\hbar\Gamma/2$ below the axis. At this point the resolvent itself has the value $R(\varepsilon_0 - \mathrm{i}a) = 1/(2\mathrm{i}a)$ and no pole.

The example uses the multiplication model of Example 12.10, not the Friedrichs model. Figure 12.3 draws its two branches.

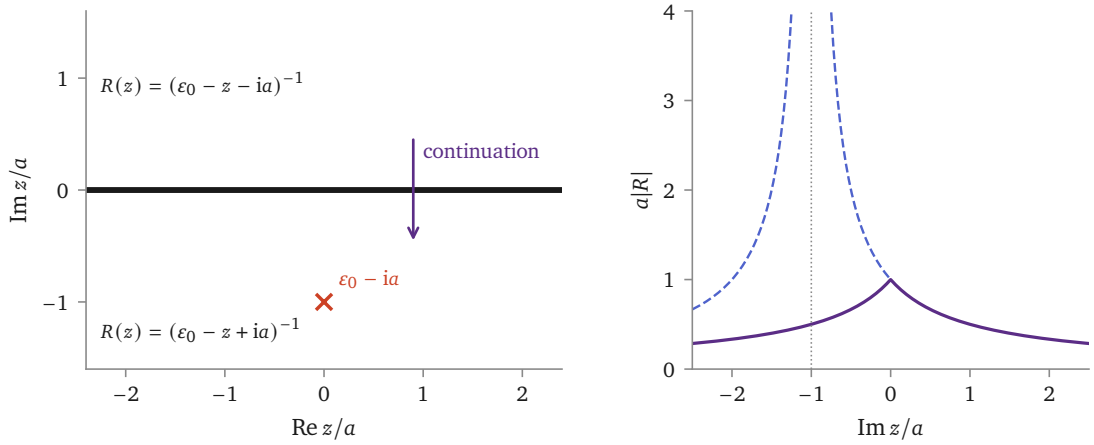


Figure 12.3: The resolvent of Example 12.14, in units with $\varepsilon_0 = 0$ and $a = \hbar\Gamma/2 = 1$. Left: the complex plane, with the formula of R in each half-plane and the real axis thick. The arrow is the continuation of the upper formula across the axis, and the cross is its pole at $-\mathrm{i}a$. Right: along the line $\operatorname{Re} z = \varepsilon_0$, the modulus of the resolvent (solid), which stays at most $1/a$, and of the continuation (dashed), which is infinite at $\operatorname{Im} z = -a$.

In the pole-and-real-axis approximation, the level's distribution of energies near the resonance is replaced by a Lorentzian about E_{res}, of half-width $\hbar\Gamma/2$, on the whole line. Example 12.10 then gives the survival $\mathcal{P}(t) = e^{-\Gamma t}$ for $t \geq 0$. Exner, p. 108, identifies Γ with *Fermi's golden rule*: $2\pi/\hbar$ times the squared coupling $|g\nu(\lambda_0)|^2$ of the level to the continuum at its own energy, times the density of continuum states there, which is 1 per unit of λ. This approximation is a model, not a consequence of Theorem 12.13. A pole is not an error bound on the decay law. Below we compute the actual density for one form factor, and Figure 12.5 compares its survival with $e^{-\Gamma t}$.

This computation uses the limit of $1/(\lambda' - z)$ as $z = \lambda + \mathrm{i}\epsilon$ approaches the real axis. Figure 12.4 draws the real and the imaginary part of this function for three values of ϵ.

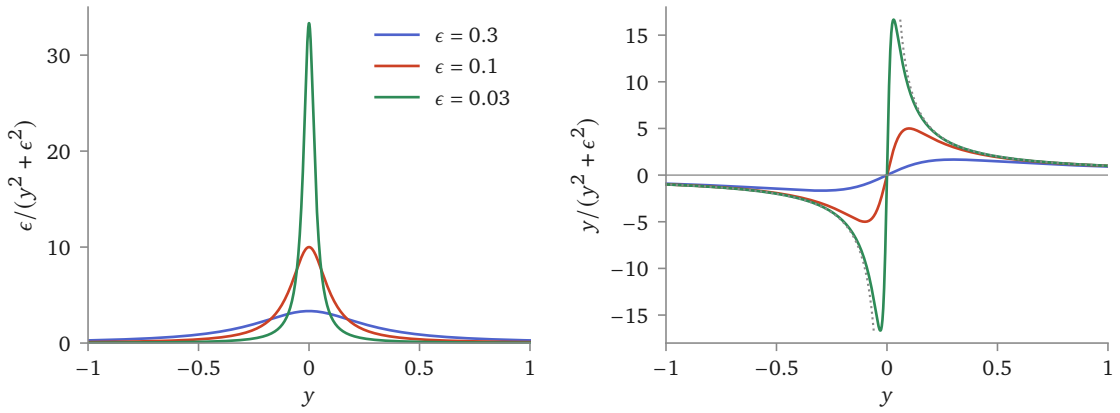


Figure 12.4: The two parts of $1/(\lambda' - z)$, with $z = \lambda + \mathrm{i}\epsilon$, against $y = \lambda' - \lambda$, for $\epsilon = 0.3$, 0.1 and 0.03. Left: the imaginary kernel $\epsilon/(y^2+\epsilon^2)$, of height $1/\epsilon$ at $y = 0$. On the whole line its area is π for every ϵ. Lemma 12.15 integrates over $y > -\lambda$ only, where the area is $\pi/2 + \arctan(\lambda/\epsilon)$, which tends to π as $\epsilon \to 0$. Right: the real kernel $y/(y^2+\epsilon^2)$, which is odd, and dotted, $1/y$. Its values at y and $-y$ are opposite, so near $y = 0$ they cancel in the integral. This cancellation gives the principal value, although the integral of $1/|y|$ near 0 is infinite.

The bracket of Proposition 12.12 has a limit on the real axis. Let $\lambda > 0$ and $\epsilon > 0$, and put $z = \lambda + \mathrm{i}\epsilon$. Then

$$\frac{1}{\lambda' - z} = \frac{\lambda' - \lambda}{(\lambda' - \lambda)^2 + \epsilon^2} + \mathrm{i}\,\frac{\epsilon}{(\lambda' - \lambda)^2 + \epsilon^2}.$$

As a function of λ', the imaginary part is π times a Lorentzian of half-width ϵ about λ (Example 12.10). As $\epsilon \to 0$ it concentrates at λ. The real part is odd about λ. This suggests the following limit.

Lemma 12.15 (Sokhotski–Plemelj formula). *Let $v \in L^2(0,\infty)$, and let $\lambda > 0$ be a point near which $|v|^2$ is continuously differentiable. Then, as $\epsilon \to 0$ with $\epsilon > 0$,*

$$\int_0^\infty \frac{|v(\lambda')|^2}{\lambda' - \lambda - \mathrm{i}\epsilon}\,\mathrm{d}\lambda' \longrightarrow J(\lambda) + \mathrm{i}\pi\,|v(\lambda)|^2,$$

where $J(\lambda)$, the principal value *of $\int_0^\infty |v(\lambda')|^2/(\lambda' - \lambda)\,\mathrm{d}\lambda'$, is the limit as $\delta \to 0$ of the integral without $(\lambda - \delta, \lambda + \delta)$.*

Proof. Put $y = \lambda' - \lambda$, and call the two parts of $1/(\lambda' - z)$ above the real kernel $y/(y^2+\epsilon^2)$ and the imaginary kernel $\epsilon/(y^2+\epsilon^2)$. Choose a with $0 < a < \lambda$ such that $|v|^2$ is continuously differentiable on $I = [\lambda - a, \lambda + a]$. By the mean value theorem, the quotient $q(y) = (|v(\lambda + y)|^2 - |v(\lambda)|^2)/y$, for $0 < |y| \le a$, is bounded, say by M.

First, let λ' lie outside I, so that $|y| \ge a$. The real kernel differs from $1/y$ by at most ϵ^2/a^3, and the imaginary kernel is at most ϵ/a^2. We multiply these bounds by $\|v\|^2 < \infty$. So the real integral outside I tends to the part of $J(\lambda)$ outside I, and the imaginary integral outside I tends to 0.

Next, take the real part on I. The real kernel is odd in y and I is symmetric about λ, so the kernel integrates the constant $|v(\lambda)|^2$ to 0. What remains is $\int_{-a}^{a} q(y)y^2/(y^2+\epsilon^2)\,\mathrm{d}y$. This differs from $\int_{-a}^{a} q(y)\,\mathrm{d}y$ by at most $M\int_{-a}^{a} \epsilon^2/(y^2+\epsilon^2)\,\mathrm{d}y \le \pi M\epsilon$. The integral of q is the part

of $J(\lambda)$ on I. Indeed, on $\delta < |y| < a$ the integrand of J is $q(y) + |v(\lambda)|^2 / y$, and the second term integrates to 0.

Finally, take the imaginary part on I. The constant $|v(\lambda)|^2$ gives $2\,|v(\lambda)|^2 \arctan(a/\epsilon)$, which tends to $\pi\,|v(\lambda)|^2$. The rest is at most $M\epsilon \int_{-a}^{a} |y| \,/ (y^2 + \epsilon^2)\,\mathrm{d}y = M\epsilon \log(1 + a^2/\epsilon^2)$, which tends to 0. The three limits add up to the formula. ∎

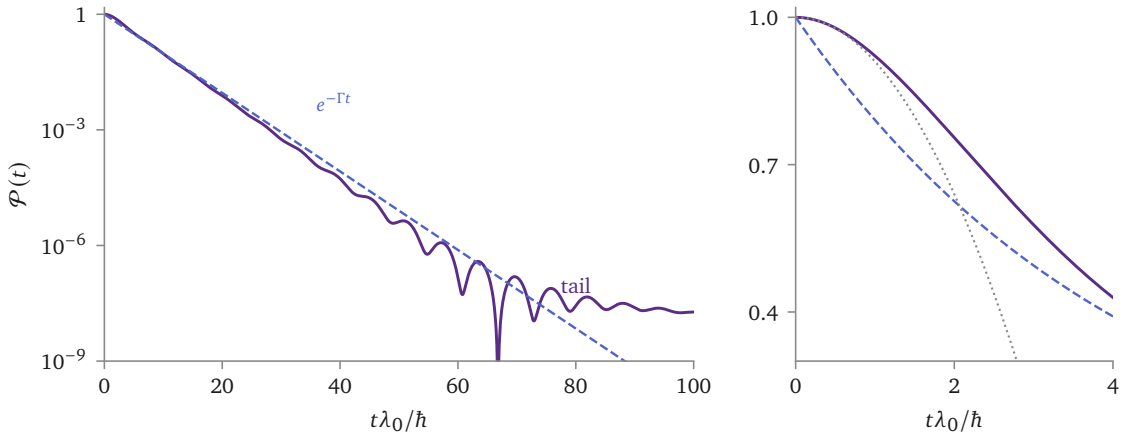


Figure 12.5: The survival probability of the level e of the Friedrichs model with $|v(\lambda)|^2 = (2/\sqrt{\pi})\lambda_0^{-3/2}\lambda^{1/2}e^{-\lambda/\lambda_0}$ and $g = 0.3\lambda_0$, computed from its distribution of energies, the density $w(E)$ derived later in this section. In units with $\hbar = \lambda_0 = 1$, the leading value $2\pi g^2\,|v(\lambda_0)|^2\,/\hbar$ of Theorem 12.13 is $\Gamma \approx 0.2347$, and $(\Delta H)^2 = g^2 = 0.09$. No error bound for the exponential is claimed. Left, on a logarithmic scale: $\mathcal{P}(t)$ and the golden-rule exponential $e^{-\Gamma t}$ (dashed). Right, short times: $\mathcal{P}(t)$, $e^{-\Gamma t}$ (dashed) and $1 - (\Delta H)^2 t^2/\hbar^2$ (dotted).

This limit suggests the leading term of Γ. The following argument is a heuristic, not a proof. Freeze the boundary value at λ_0 for all z near λ_0. The bracket becomes $\lambda_0 - z - g^2 J(\lambda_0) - \mathrm{i}\pi g^2\,|v(\lambda_0)|^2$, and its zero is

$$z = \lambda_0 - g^2 J(\lambda_0) - \mathrm{i}\pi g^2\,|v(\lambda_0)|^2 .$$

It lies below the real axis, at the distance $\pi g^2\,|v(\lambda_0)|^2$. By the theorem, $\hbar\Gamma/2 = \pi g^2\,|v(\lambda_0)|^2 + O(g^4)$, so this distance is only the leading order. Boundary values on the real axis also do not show that this zero is a pole of the continued function, or that the pole dominates the evolution in time.

The same limit gives the level's distribution of energies, and this step is not a heuristic. By the spectral theorem, the function $z \mapsto \langle e, (H_g - z)^{-1} e\rangle$ is $\int (E - z)^{-1}\,\mathrm{d}\mu_e(E)$. Outside a set of measure zero (§ 0.4), the density of the absolutely continuous part of μ_e is $1/\pi$ times the limit of the imaginary part of this function at $E + \mathrm{i}\epsilon$ (Teschl, Thm 3.23, cited). Let $g \neq 0$, and let $|v|^2$ be continuously differentiable on $(0, \infty)$, as it is under the hypotheses of Theorem 12.13 and for the form factor below. Then Proposition 12.12 and the limit above give this density where $v(E) \neq 0$:

$$w(E) = \frac{g^2\,|v(E)|^2}{(\lambda_0 - E - g^2 J(E))^2 + \pi^2 g^4\,|v(E)|^4} .$$

For $E < 0$, we have $\lambda - E \geq |E|$ for every $\lambda \geq 0$. So the bracket tends to the real number $\lambda_0 - E - g^2 \int_0^\infty |v(\lambda)|^2\,(\lambda - E)^{-1}\,\mathrm{d}\lambda$. Indeed, at $y = \lambda - E$ the real kernel differs from $1/y$

by at most $\epsilon^2/|E|^3$, and the imaginary kernel is at most ϵ/E^2. We multiply these bounds by $g^2\,\|v\|^2 < \infty$, and the term $-\mathrm{i}\epsilon$ of $\lambda_0 - z$ also tends to 0. The number $\langle e, (H_g - E - \mathrm{i}\epsilon)^{-1} e\rangle$ is the reciprocal of the bracket, so its imaginary part tends to 0 wherever this real limit is not 0. So the density is 0 there. This real limit is strictly decreasing in E, since $-E$ and $-1/(\lambda - E)$ decrease as E grows. So it vanishes at one point $E < 0$ at most, and changing a density at one point changes no probability. We put $w(E) = 0$ for $E < 0$. That point, if it exists, is a negative eigenvalue of H_g, with eigenvector $\{1, -gv/(\lambda - E)\}$, and it adds a point weight separately. Near the resonance, freezing $|v|^2$ and J at λ_0 turns w into the Lorentzian of Example 12.10 about $\lambda_0 - g^2 J(\lambda_0)$, of half-width $\pi g^2\,|v(\lambda_0)|^2$. The centre E_{res} of Theorem 12.13 agrees with $\lambda_0 - g^2 J(\lambda_0)$ to the order g^2 (Exner, (2.8a)). This is a shape near the peak, not the exact density, since w vanishes for $E < 0$. By Example 12.10, a Lorentzian on the whole line gives an exact exponential. The threshold at $E = 0$ and the difference between w and the Lorentzian produce the departures from it.

Take the form factor $|v(\lambda)|^2 = (2/\sqrt{\pi})\lambda_0^{-3/2}\lambda^{1/2}e^{-\lambda/\lambda_0}$, with $\|v\| = 1$, and $g = 0.3\lambda_0$. In units with $\hbar = \lambda_0 = 1$, it has these numbers:

Leading value of Γ	0.2347
Centre of the frozen Lorentzian, $\lambda_0 - g^2 J(\lambda_0)$	1.0137
Half-width of the frozen Lorentzian, $\pi g^2\,\lvert v(\lambda_0)\rvert^2$	0.1174
Pole of the continued function	1.0243 – 0.1261i
Peak of w	1.0190

The exact pole lies 0.1261 below the real axis, against the leading 0.1174. The number 0.1261 is not a half-width of the curve w, which is not symmetric about its peak. Figure 12.6 shows the pole and the density.

For this form factor and $g = 0.3\lambda_0$, the level's distribution is $w(E)\,\mathrm{d}E$, which Figure 12.5 uses. The rest of μ_e, its point part and its singular continuous part, gives probability 0 outside the set of real E at which the imaginary part of $\langle e, (H_g - E - \mathrm{i}\epsilon)^{-1} e\rangle$ is unbounded as $\epsilon \to 0$ (Teschl, (3.77) and Cor. 3.24, cited). We show that this set is empty. By Proposition 12.12, the number $\langle e, (H_g - z)^{-1} e\rangle$ at $z = E + \mathrm{i}\epsilon$ is the reciprocal of the bracket. Let $E \le 0$ and $\epsilon > 0$. Since $(\lambda - E)/((\lambda - E)^2 + \epsilon^2) \le 1/\lambda$, the real part of the bracket is at least $\lambda_0 - g^2 \int_0^\infty |v(\lambda)|^2\,\lambda^{-1}\,\mathrm{d}\lambda = 0.82\lambda_0 > 0$. So the reciprocal has modulus at most $1/(0.82\lambda_0)$. Now let $E > 0$. By the limit above, the bracket tends to a number with the imaginary part $-\pi g^2\,|v(E)|^2 < 0$, since this form factor is positive on $(0, \infty)$. So the reciprocal has a finite limit. Hence μ_e has only its absolutely continuous part. Its density is w on $(0, \infty)$ and 0 for $E < 0$, since the limit of the bracket is not 0 there. Numerically, $\int_0^\infty w\,\mathrm{d}E = 1$, as this requires.

Figure 12.5 computes $\mathcal{P}(t)$ from w, since the survival amplitude of a state whose distribution has the density w is $\int e^{-\mathrm{i}tE/\hbar} w(E)\,\mathrm{d}E$ (§ 12.2). The decay starts flat, as Proposition 12.8 requires, follows $e^{-\Gamma t}$, with the leading Γ, to within a factor of 2 for about nine lifetimes, and ends in a tail that falls like t^{-3}, slower than every exponential, as Theorem 12.9 requires. Near the threshold $E = 0$, $|v(E)|^2$ grows like $E^{1/2}$, and the denominator of w tends to $(0.82\lambda_0)^2$, so $w(E)$ grows like a constant times $E^{1/2}$. For this density, the square-root edge at 0 accounts for the observed power-law tail. It gives a survival amplitude of the order of $t^{-3/2}$ at late times, and so the tail t^{-3}. Where the exponential and the tail are of the same size, the two contributions to the amplitude interfere, and $\mathcal{P}(t)$ oscillates with a period close to $2\pi\hbar/E_{\mathrm{res}}$.

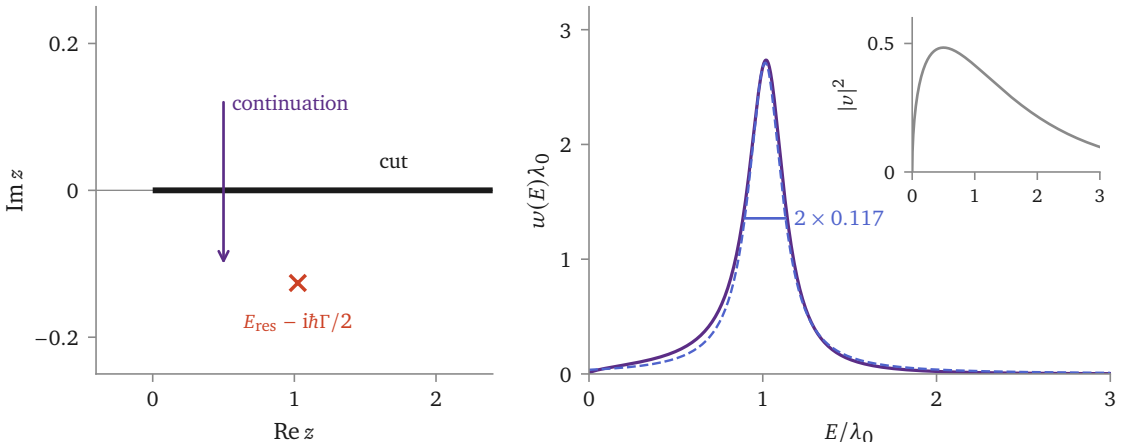


Figure 12.6: The form factor of the text, with $\hbar = \lambda_0 = 1$ and $g = 0.3$. Left: the complex plane, with the cut $(0, \infty)$ thick. The arrow is the continuation from the upper half-plane across the cut, and the cross is the pole at about 1.0243 – 0.1261i of the continued function, not a pole of the resolvent of H_g. Right: the exact density $w(E)$ (solid), and the Lorentzian with $|v|^2$ and J frozen (dashed), with the leading half-width 0.1174; the bar marks its full width. Inset: $|v|^2$ on its own scale.

Self-check 12.5

1. For the form factor of Figure 12.5, show that $\Gamma = 4\sqrt{\pi}\, e^{-1} g^2/\hbar\lambda_0$ to the order g^2, and compare the lifetime $1/\Gamma$ with $\hbar/\Delta H$ at $g = 0.3\lambda_0$.
2. Besides $v \in L^2(0, \infty)$, what does Theorem 12.13 assume about v and about g? Does the pole it gives bound $\left|\mathcal{P}(t) - e^{-\Gamma t}\right|$ for all $t > 0$?

The physics

The $2p$ level of hydrogen decays to $1s$ by emitting a photon of the Lyman α line, of energy 10.2 eV. The atom in $2p$ plays the role of e, and the atom in $1s$ with one photon of any energy and direction the continuum. The golden rule, applied to the coupling of the electron to the electromagnetic field, which this course does not treat, gives $\Gamma = (2/3)^8 \alpha_{\text{fs}}^5 mc^2/\hbar \approx 6.27 \times 10^8\,\text{s}^{-1}$, with $\alpha_{\text{fs}} \approx 1/137$ the fine-structure constant: a lifetime of 1.60 ns. In 1966 Bickel and Goodman passed a fast beam of hydrogen molecular ions, H_3^+, through a thin carbon foil, which breaks them up into excited atoms, and recorded the Lyman α light at increasing distances behind the foil. They measured 1.600 ± 0.004 ns. The width $\hbar\Gamma \approx 4.1 \times 10^{-7}$ eV is about 4×10^{-8} of the line's energy: the level is very narrow.

Decay

States: every state is $\psi_{\text{pp}} + \psi_{\text{c}}$, and the evolution keeps both parts (Proposition 12.2). For a potential, a state of $\mathcal{H}_{\text{pp}}$ stays in a bounded region for all times, up to an arbitrarily small probability, a state $\Omega_{\text{out}}\chi$ leaves every bounded region as $t \to \infty$ (Theorem 12.6), and by the cited Enss theorem every state of $\mathcal{H}_{\text{c}}$ leaves (Corollary 12.7). *Evolution:* for a strongly continuous unitary group with the Hamiltonian H and a unit vector

$\psi \in D(H)$, $\mathcal{P}(t) = 1 - (\Delta H)^2 t^2/\hbar^2 + o(t^2)$ (Proposition 12.8), so no state of finite energy spread decays exactly exponentially; if H is bounded below, no $\mathcal{P}$ is bounded by $Ce^{-\beta t}$ with $\beta > 0$ for all $t > 0$ (Theorem 12.9, cited). *Measurement:* for a potential, by the cited Enss theorem, at late times a reading of the position in $[-L, L]$ sees only the point part: its probability differs from $\left\| 1_{[-L,L]} U_t \psi_{\mathrm{pp}} \right\|^2$ by an amount that tends to 0.

Summary.

- *The point part*: the closure of the sums of eigenvectors and its orthogonal complement are kept by the evolution; for a unit vector of the point part the survival amplitude is $\sum_E w_E e^{-\mathrm{i}Et/\hbar}$, with mean survival probability $\sum_E w_E^2 > 0$ (Proposition 12.2).
- *States that leave*: every free state leaves every bounded region (Proposition 12.4); for a potential, the states of $\mathcal{H}_{\mathrm{pp}}$ stay, up to small tails uniformly in time, and the states $\Omega_{\mathrm{out}}\chi$ leave (Theorem 12.6); with Enss's theorem, all of $\mathcal{H}_{\mathrm{c}}$ leaves (Corollary 12.7, sketched).
- *Short times*: for every strongly continuous unitary group, $\mathcal{P}(t) = 1 - (\Delta H)^2 t^2/\hbar^2 + o(t^2)$ for a unit vector $\psi \in D(H)$ (Proposition 12.8); Khalfin's theorem for H bounded below is cited.
- *An exact exponential*: the Lorentzian state of multiplication by E has $\mathcal{P}(t) = e^{-\Gamma|t|}$; it lies outside $D(H)$, and H is not bounded below (Example 12.10).
- *The golden rule*: in the Friedrichs model the level's resolvent is explicit (Proposition 12.12). Under the analytic-extension and nonzero-coupling hypotheses of Theorem 12.13, for sufficiently small nonzero g, its resonance has $\Gamma = 2\pi g^2 |v(\lambda_0)|^2/\hbar + O(g^4)$ (Theorem 12.13, cited).

The $2p$ level of hydrogen opened this lecture, and the course ends with hydrogen, as Lecture 0 announced. Lecture 13 takes the particle into space, where rotations act and sort the eigenvectors into multiplets. Lecture 14 builds the Hamiltonian of hydrogen, whose Coulomb potential is singular at the proton, and Lecture 15 finds its levels and their multiplicities.

Exercises. Exercise Sheet 12.

Self-check answers.

12.1 1. With $E_2 = 4E_1$, $\langle \psi, U_t \psi \rangle = (e^{-\mathrm{i}E_1 t/\hbar} + e^{-4\mathrm{i}E_1 t/\hbar})/2$, so $\mathcal{P}(t) = \cos^2(3E_1 t/2\hbar)$. Its mean is $\frac{1}{2} = w_{E_1}^2 + w_{E_2}^2$, and $\mathcal{P}(t) = 1$ at the times $t = 2\pi\hbar k/3E_1$, $k \in \mathbb{Z}$.

12.2 1. For $t \neq 0$, $\langle \psi, U_t \psi \rangle = \int_0^1 e^{-\mathrm{i}tx/\hbar}\, \mathrm{d}x = (\hbar/\mathrm{i}t)(1 - e^{-\mathrm{i}t/\hbar})$, and $\left|1 - e^{-\mathrm{i}t/\hbar}\right| \leq 2$.

12.3 1. By Postulate 5.6, each yes leaves the state ψ, up to a phase, which then evolves for the time t/n; so the answers are yes with the probability $\mathcal{P}(t/n)$ each, and all n with $\mathcal{P}(t/n)^n$. By Proposition 12.8, $n \log \mathcal{P}(t/n) = n \log(1 - (\Delta H)^2 t^2/\hbar^2 n^2 + o(1/n^2)) \to 0$. So frequent tests keep the tested system in the ray of ψ, with a probability that tends to 1: the *Zeno effect*.

12.4 1. At $E = \varepsilon_0 \pm \hbar\Gamma/2$ the denominator is $2(\hbar\Gamma/2)^2$, twice its value at ε_0. For $\Gamma = 10^9\,\mathrm{s}^{-1}$ the half-width is $\hbar\Gamma/2 \approx 3.3 \times 10^{-7}$ eV.

12.5 1. $|v(\lambda_0)|^2 = (2/\sqrt{\pi})e^{-1}/\lambda_0$, so $\Gamma = 2\pi g^2 |v(\lambda_0)|^2/\hbar = 4\sqrt{\pi}\, e^{-1} g^2/\hbar\lambda_0$. At $g = 0.3\lambda_0$, $\Gamma \approx 0.235\lambda_0/\hbar$ and $1/\Gamma \approx 4.3\hbar/\lambda_0$, while $\Delta H = |g|\,\|v\| = 0.3\lambda_0$ and $\hbar/\Delta H \approx 3.3\hbar/\lambda_0$: the two times are of the same order. 2. It assumes that $|v|^2$ extends from $(0, \infty)$ to a meromorphic function on an open set of complex numbers that contains $(0, \infty)$, that $v(\lambda_0) \neq 0$, and that g is real, nonzero and small enough. The pole gives no such bound. The theorem locates a pole of the continued function and gives Γ to the order g^2. By Proposition 12.8, $\mathcal{P}(t)$ starts flat, unlike $e^{-\Gamma t}$. By Theorem 12.9, applied to the H_g bounded below, $\mathcal{P}(t)e^{\Gamma t}$ is unbounded at late times. Figure 12.5 compares the two curves numerically for one form factor.

References

Teschl, *Mathematical Methods in Quantum Mechanics*, (3.77) (p. 103), Cor. 3.24 (p. 109), Lemma 3.19 (p. 105), Thm 3.23 (p. 109), p. 112, Thms 5.6–5.7 (pp. 128–129), Thm 12.11 (p. 255), Thm 12.12 (p. 256). Amrein, *Hilbert Space Methods in Quantum Mechanics*, Prop. 5.18 (p. 222). Teta, *A Mathematical Primer on Quantum Mechanics*, Prop. 6.2 (p. 147). Exner, *Open Quantum Systems and Feynman Integrals*, Lemma 1.5.5 (p. 25), Thm 1.5.7 (p. 27), §3.1 (p. 103), Thm 3.2.3 (p. 107), p. 108. Bickel and Goodman, *Phys. Rev.* 148, 1 (1966).